

Romeo Silvestri

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Summary

Applied Machine Learning Engineer with **3+ years of work experience** and a strong foundation in statistics and data science. Specializing in AI research, MLOps workflows, reinforcement learning and Generative AI, I design and deploy robust, scalable models. Author of peer-reviewed papers, eager to apply my technical expertise and research to deliver projects with real-world impact.

Skills

Languages & Frameworks: Python, R, SQL, FastAPI, Gymnasium, Numpy, Optuna, Pandas, PySpark, PyTorch, Scikit-learn, TensorFlow

Tools & Databases: Airflow, Docker, Git, LangChain, LangGraph, MLflow, Power BI, Prefect, Spark, MongoDB, MySQL, PostgreSQL

Techniques: A/B Testing, Agentic AI, Classification, Clustering, Deep Learning, Generative AI, Machine Learning, Model Monitoring, Natural Language Processing (NLP), Neural Networks, RAG, Reinforcement Learning, Regression, Time Series Forecasting

Work Experience

ML Engineer | Fondazione Bruno Kessler (FBK), Trento 12/2023 - Present

- Optimized ML pipelines and provided APIs in a research and production environment using **Python**, **FastAPI**, **GitLab**, **LangGraph**, and **MLflow**, while collaborating closely with domain experts to address complex problems and streamline processes.
- Designed, tested, and deployed a Decision Support System (DSS) for irrigation, integrating heterogeneous data sources, including IoT sensors and weather data, reducing operational costs by around **50%** across several companies.
- Developed a Deep Reinforcement Learning agent with PPO optimization, leveraging a digital twin environment using a surrogate and a physical model for simulation and policy training, with synthetic weather data generation via KNN bootstrap.
- Developed a multi-agent LLM-based chatbot for the European AI Factories project, leveraging pre-trained models and Retrieval-Augmented Generation (RAG) to support decision-making. The system provides real-time, data-driven advice on agriculture, integrating legal guidelines and technical best practices, with support for global and region-specific recommendation models.
- Built state-of-the-art time series forecasting models for sensor trends using ARIMA, XGBoost, and LSTM neural networks, achieving around **5%** prediction error through advanced feature engineering, hyperparameter tuning, and rigorous validation.

Data Scientist | Youbidata, Bologna 10/2023 - 11/2023

- Developed a centralized financial monitoring dashboard using **PostgreSQL** and **Power BI** to provide KPIs for a consulting client.

Statistician (Internship) | ISTAT, Rome 03/2023 - 08/2023

- Analyzed national demographic data using **Excel**, **R** and **SQL** to uncover socio-economic trends and produce periodic reports.
- Evaluated the Austrian quality framework's suitability for the Italian Census, achieving statistically significant results.

Education

M.Sc. in Statistics & Data Science | Sapienza University of Rome 11/2020 - 03/2023

Thesis: *Analyzing Forecasting Models for Real Estate Markets with a Case Study of Madrid Prices* Graduation Grade: 107/110

B.Sc. in Statistics | University of Padua 10/2017 - 11/2020

Thesis: *The Impact of Migration on Population Dynamics in Italian Provinces* Graduation Grade: 102/110

Publications

A Generalizable Retrieval-Augmented LLM Framework for PRISMA-Aligned Systematic Reviews ESWA (submitted)

Deep Reinforcement Learning for irrigation optimization using a digital twin of crop-soil dynamics IEEE MetroInd 2026 (EA accepted)

Smart Irrigation with Fuzzy Decision Support Systems in Trentino Vineyards MDPI Sensors

A Fuzzy Decision Support System to Optimize Irrigation Practices in Trentino Region CoDIT 2025

Comparative Analysis of Soil Moisture Interpolation Techniques in Apple Orchards of Trentino Region IEEE MetroAgriFor 2024

Additional Information

Certifications: Deep Learning (DeepLearning.AI), Google Data Analytics, IBM Data Science

Teaching: Private Tutor in Statistics and IT (2020-2021), University Tutor at Sapienza University (2022)

Organizations: Italian Volunteers Blood Donation (AVIS), Mensa High IQ Society

Interests: Art, Chess, Mixed Martial Arts, Music, Poetry, Technology, Traveling

I authorize the processing of personal data contained in my curriculum vitae based on art. 13 GDPR 679/16.